

PH225 - Quantum Mechanics

Dirac notation of Linear Algebra.

* Linear Vector Spaces: (V)

Vector: $|v\rangle = | \rangle = \text{KET}$

Standard vector $\vec{v}$ = has magnitude and direction

But, set of these is also a vector.

✓ V consists of a set of vectors $|v\rangle$

Say $\exists |v\rangle, |w\rangle \in V$ are two vectors

Then, ① $|v\rangle + |w\rangle \in V$ (sum should be defined)

② $a = \text{number}$ and $|v\rangle \in V$, $a \cdot |v\rangle \in V$

Linear:

(product): $a(|v\rangle + |w\rangle) = a \cdot |v\rangle + a \cdot |w\rangle$

$$(a+b)(|v\rangle) = a \cdot |v\rangle + b \cdot |v\rangle$$

$$|v\rangle + (|w\rangle + |z\rangle) = (|v\rangle + |w\rangle) + |z\rangle$$

(associativity)

i) $\exists |0\rangle$ such that, $|v\rangle + |0\rangle = |0\rangle + |v\rangle = |v\rangle$

ii) $\exists |-v\rangle$ such that, $|v\rangle + |-v\rangle = |-v\rangle + |v\rangle = |0\rangle$

iii) $|v\rangle + |w\rangle = |w\rangle + |v\rangle$ (commutative)

i, ii, iii, iv $\Rightarrow$ These all together make (in groups)

Basis Vectors:

Linear Independence:

$$a_v |v\rangle + a_w |w\rangle = 0$$

If $a_v = a_w = 0$ then $|v\rangle, |w\rangle$ are linearly independent vectors

o/w they are dependent.

✓ The above comparison can be done for any number of vectors.
But what is the max number?

Dimension of Vector space = max # of Linearly Independent vectors.

dimension of $\begin{bmatrix} a & b \\ c & d \end{bmatrix} = 4$
(2x2 matrix)

dim of 2x2 symmetric matrices = 3

Basis of vector space =

Consider V [consisting of $|u\rangle$]

let $\mathcal{V}$ be a conjugate vector space $[CV]$ consisting of $\langle u|$

Then $\mathcal{V}$ is one-to-one mapping b/w $\langle u|$ and CV

$\langle u| + \langle v| = \langle u+v| = \text{BRACKET} \Rightarrow$ Dirac's Bracket notation.

Ex: if $\langle A| = \begin{bmatrix} A_1 & A_2 & A_3 \end{bmatrix}$ then $\langle A| = [A_1 \ A_2 \ A_3]$

⊛ We cannot take products of KET and KET

For every BRA $\exists$ KET and viceversa.

* Inner Product:

$|u\rangle, |v\rangle$ = both are KETS

Inner product $\langle u|u\rangle$

here we took

BRA of $|u\rangle$

this is KET form $|u\rangle$

Inner product needs to be (definite: $\langle u|u \rangle > 0$)

(i) Positive semidefinite: $\langle u|u \rangle \geq 0$

(ii) Linearity: $\langle u|(w\rangle + |z\rangle) = \langle u|w\rangle + \langle u|z\rangle$

(iii) $\langle w|u \rangle = \langle u|w \rangle^*$

should be complex conjugates of each other.

(*) Converting KET to BRA

$|u\rangle \in V \longrightarrow \exists \langle u| \in CV$

Take $\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$ adjoint operation $\longrightarrow$ Transpose $\langle u| = [u_1^* \ u_2^* \ \dots \ u_n^*]$

(*) Normalised vectors:

$|u\rangle \neq |0\rangle \quad |\hat{u}\rangle = \frac{|u\rangle}{\sqrt{\langle u|u \rangle}} \quad \langle \hat{u}|\hat{u} \rangle = 1$

Orthogonal vectors:

If $\langle u|w \rangle = 0$ then $|u\rangle, |w\rangle$ are orthogonal to each other

(*) Orthogonal Basis

$|u_1\rangle, |u_2\rangle, \dots, |u_n\rangle$ are linearly independent

if $\sum_{i=1}^n a_i |u_i\rangle = 0$ only if $a_i = 0 \ \forall i$

Then $|u_i\rangle \ \forall i$ becomes basis V and maximum linearly independent vectors in this V

If these $|u_i\rangle$ are orthogonal to each other $\Rightarrow$ called an orthogonal basis

* Orthonormal Basis:

Max. Vectors which are orthogonal to each other and have $\langle u|u \rangle = 1$

Converting Basis into Orthonormal Basis:

$$|1\rangle = \frac{|u_1\rangle}{\sqrt{\langle u_1|u_1\rangle}} \Rightarrow \langle u_1|u_1\rangle = 1$$

$$|u_2'\rangle = |u_2\rangle - \langle 1|u_2\rangle \cdot |1\rangle$$

$$|2\rangle = \frac{|u_2'\rangle}{\sqrt{\langle u_2'|u_2'\rangle}} \Rightarrow \langle 2|2\rangle = 1$$

$$|u_3'\rangle = |u_3\rangle - \langle 1|u_3\rangle \cdot |1\rangle - \langle 2|u_3\rangle \cdot |2\rangle$$

and this process continues till we get

$$|1\rangle, |2\rangle, \dots, |n\rangle$$

The above method is called as Gram-Schmidt Procedure.

Kronecker delta (δ_{ij})

$$\langle i|j\rangle = \delta_{ij} = \begin{cases} 1 & \text{for } i=j \\ 0 & \text{for } i \neq j \end{cases}$$

represents Identity matrix

* Expressing any vector in terms of basis vectors

$$|u\rangle = |u_1\rangle + |u_2\rangle + \dots + |u_n\rangle$$

$$|u\rangle = \sum_{i=1}^n w_i |u_i\rangle \Rightarrow \text{finding these weights } w_i$$

✓ Convert into orthonormal basis first

$$|1\rangle, |2\rangle, \dots, |n\rangle$$

$$|u\rangle = \sum w_i |i\rangle$$

$$\langle j | \omega \rangle = \sum_i \omega_i \langle j | i \rangle = \omega_j \langle j | j \rangle = \omega_j$$

ie $\forall i \left[\omega_i = \langle i | \omega \rangle \right]$ $\begin{matrix} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{matrix}$ $\langle j | i \rangle = \delta_{ji}$

②*

$$|z\rangle = \sum_i z_i |i\rangle$$

$$\begin{bmatrix} \langle i | z \rangle \\ \vdots \\ \langle j | z \rangle \end{bmatrix} = \langle i |$$

$$\langle i | \sum_j z_j |j\rangle = \langle i |$$

vector used

$$x \text{ matrix } n \times n = \Omega$$

$$\Lambda \text{ rotation matrix } \text{bno } \langle i | \Omega = \langle i | \Omega$$

$$\langle i | \Lambda \Omega = \langle i | \Omega = \langle i | \Lambda$$

③ Outer Product:

$$|u\rangle \langle w| = \text{outer} \quad \langle u | \Omega | v \rangle = \left[\langle u | \langle \omega | = \left[\begin{matrix} \vdots \\ \langle u | \omega \rangle \end{matrix} \right] \right]$$

= square matrix

Linear operator: $\Omega \mapsto$ called as tensor in classical Mech.

$$\Omega |u\rangle = \sum_i \Omega_{ij} |i\rangle \quad \Omega = n \times n \text{ matrix}$$

operator

re presented using column matrix. $(\Omega \quad \Lambda \quad \Lambda \quad \Omega)$

(n dimensional)

$$|u\rangle \langle u| = \Omega |u\rangle \quad \langle u | \Omega = \langle u | \Omega$$

$$|u\rangle = \sum_i u_i |i\rangle \quad (\text{after choosing orthonormal basis})$$

This $\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$ is unique representation for $|u\rangle$ for fixed basis.

How is Ω represented?

Taking projections of $|u\rangle$ on $|u'\rangle$

$$\langle i | \Omega | j \rangle = \text{scalar} = \langle i | \Omega | j \rangle$$

$$\langle i | \Omega | j \rangle = \begin{bmatrix} \langle i | \Omega | 1 \rangle & \langle i | \Omega | 2 \rangle & \dots & \langle i | \Omega | n \rangle \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

⊛ Consider operator $\Omega, |u\rangle \in V$

$$|u\rangle = \sum_i u_i |i\rangle$$

basis vector

$$|u\rangle = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$$

$\Omega = n \times n$ matrix

• Consider $\Omega |u\rangle = |u'\rangle$ and new operator Λ

$$\Lambda |u'\rangle = |u''\rangle = \Omega \Lambda |u\rangle$$

and

$$\Lambda |u\rangle = |\tilde{u}\rangle$$

$$\langle \tilde{u} | \Omega | u \rangle = \langle \tilde{u} | u' \rangle = \langle \Lambda \Omega | u \rangle$$

$$(\Omega \Lambda - \Lambda \Omega) |u\rangle \neq 0 \Rightarrow \boxed{\Omega \Lambda - \Lambda \Omega \neq 0}$$

⊛ Operators don't commute.

⊛ $(\Omega \Lambda - \Lambda \Omega) =$ known as commutator $= [\Lambda, \Omega]$

$$\Omega |u\rangle = |u'\rangle \Rightarrow \langle u' | \Omega^\dagger = \langle u' |$$

$\Omega^\dagger =$ adjoint of Ω

called as Ω dagger

$$\boxed{\Omega^\dagger = (\Omega^*)^T}$$

$$\Omega [\alpha |u\rangle + \beta |u'\rangle] = \alpha \Omega |u\rangle + \beta \Omega |u'\rangle$$

These are called linear operators.

⊛ These are called linear operators.

Three kinds of linear operators: For real nos $\Rightarrow$ no need complex conjugation
 ① Hermitian Operator: $\Omega^\dagger = \Omega$ (adjoint = itself)

② Anti Hermitian Operator:

$$\Omega^\dagger = -\Omega$$

③ Unitary Operator: (U)

$$U^\dagger U = I = U U^\dagger \quad (\text{dagger} = \text{its inverse})$$

④ In QM, we take about operator which can be inverted.
 ie if $\Omega|u\rangle = |v\rangle$ then $\exists \Omega^{-1}$ st $\Omega^{-1}|v\rangle = |u\rangle$
 where $\Omega \Omega^{-1} = \hat{I} = \Omega^{-1} \Omega$

$$\langle v | \hat{I} | u \rangle = \langle v | u \rangle \Rightarrow \hat{I} = \sum_i |i\rangle \langle i|$$

⑤ Actions of operator is also associative.

$$\Gamma(\Lambda \Omega) |u\rangle = (\Gamma \Lambda) \Omega |u\rangle$$

Groups $\rightarrow$ continuous groups (Sophisticated)
 discrete groups (1930) $\Rightarrow$ to solve a general polynomials (Galois)

Operators in QM are Lie groups

* Given a vector space V , $\{|i\rangle\}$ = orthonormal basis

$$|u\rangle = \sum_i u_i |i\rangle \rightarrow \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad \langle i | u \rangle = u_i$$

How does operator Ω look in this basis?

$$\Omega_{ij} = \langle i | \Omega | j \rangle \quad (\text{elements in operator of } \Omega)$$

$\underbrace{\langle i | \Omega | j \rangle}_{\text{one vector}} \rightarrow \text{dot product}$

$$|i'\rangle = U|i\rangle$$

$$\langle i'| = \langle i|U^\dagger = \langle i| \left(\sum_j U_{ji} |j\rangle \right) = \sum_j \langle i|U_{ji} |j\rangle$$

$$= \sum_j U_{ji} \langle i|j\rangle$$

$$\boxed{\langle i'| = \sum_j U_{ji} \langle i|j\rangle}$$

$$\langle i'|U|i\rangle = \sum_j \langle i'|U_{ji} |j\rangle = \sum_j \langle i|U_{ji} |j\rangle$$

$$\langle i'|U|i\rangle = \sum_j \langle i'|U_{ji} |j\rangle = \sum_j U_{ji} \langle i'|j\rangle$$

$$\sum_j U_{ji} \langle i'|j\rangle$$

$$= \sum_j \langle i'|U_{ji} |j\rangle = \sum_j U_{ji} \langle i'|j\rangle$$

$$\boxed{\langle i'|U|i\rangle = \sum_j U_{ji} \langle i'|j\rangle}$$

$$\boxed{U_{ji} = \sum_k U_{kj} \langle i'|k\rangle = \sum_k U_{kj} U_{ki}}$$

Converting U : (vs basis conversion)

$$|i'\rangle: U|i\rangle \rightarrow |i'\rangle$$

$$|j'\rangle: U|j\rangle \rightarrow |j'\rangle$$

$$\boxed{\langle i'|U|j'\rangle = \langle i|U^\dagger U|j\rangle}$$

Active transformation
(change basis).

Passive transformation

(retains basis, changing operators)

④ Given operator Ω , $\langle \psi | \psi \rangle \in \mathbb{R}$ $\langle \psi | \psi \rangle = \Omega$

if $\Omega |\psi\rangle = k \cdot |\psi\rangle$ then $|\psi\rangle =$ eigen vector
 $\downarrow$
 number of Ω with $k =$ eigen value

Prove:

① Hermitian operators have real eigen values.

② Two eigenvectors of a Hermitian operator with different eigen values, must be orthogonal.

③ All operators in Quantum Mechanics are Hermitian.

④ Degeneracy = two/more eigenvectors of same Ω having same eigen values.

ie

$$\Omega |\psi_1\rangle = \omega |\psi_1\rangle$$

$$\Omega |\psi_2\rangle = \omega |\psi_2\rangle$$

} diff eigenvector

$\downarrow$
 linearly Independent

$\Omega =$ Hermitian.

$$\Omega (\alpha |\psi_1\rangle + \beta |\psi_2\rangle) = \omega (\alpha |\psi_1\rangle + \beta |\psi_2\rangle)$$

$\Rightarrow$ ∞ -no. of eigen vectors are present in degeneracy case.

⑤ So, we assume we are not degenerate.

Ex:

$$\Omega = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

characteristic eqn $= 0$.

$\Rightarrow$ we get eigen values = 0, 2, 2

then eigen value = 2 is the degenerate space

Finding eigen vector for $\omega = 0$.

$$(\Omega - \omega I) \psi = 0 \Rightarrow \psi = U U^\dagger U$$

$$\Omega \cdot \cdot |w\rangle = 0.1w\rangle$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

we get $|w\rangle = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

Now, finding eigen vector for $w=2$,

$$\begin{bmatrix} 1-2 & 0 & 1 \\ 0 & 2-2 & 1 \\ 1 & 0 & 1-2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$x_1 = x_3$$

$$x_3 = 0$$

$$x_1 = x_3$$

$$\Rightarrow x_1 = 0, \quad x_2, \quad x_3 = 0$$

free variable $\Rightarrow \infty$ choices.

For $w=2$, two orthogonal vectors

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} = \Omega$$

* Unitary Operators:

Theorem: Eigen values of unitary operators are complex nos of unit modulus.

Eigen vectors of unitary operators are mutually orthogonal (assuming no degeneracy)

$$U^\dagger U = U U^\dagger = I \quad (\text{for unitary operators})$$

Say, $U =$ unitary operator,
 $U|\omega_1\rangle = \omega_1|\omega_1\rangle \quad U|\omega_2\rangle = \omega_2|\omega_2\rangle$
 $\langle\omega_2|U^\dagger \cdot U|\omega_1\rangle = \omega_1\omega_2^* \cdot \langle\omega_2|\omega_1\rangle$

$$(U|\omega_2\rangle)^\dagger = U^\dagger$$

$$U^\dagger \cdot U = I$$

$$\Rightarrow \langle\omega_2|\omega_1\rangle = \omega_1\omega_2^* \langle\omega_2|\omega_1\rangle$$

$$\Rightarrow \underbrace{(\omega_1\omega_2^* - 1)}_{=0} \underbrace{(\langle\omega_2|\omega_1\rangle)}_{\neq 0} = 0$$

* Simultaneous diagonalization of Hermitian Operators:

Say, $\Omega, \Lambda \rightarrow$ Hermitian

$$[\Omega, \Lambda] = 0$$

Non-degenerate

$$\Omega|\omega_i\rangle = \omega_i|\omega_i\rangle$$

$$\Omega\Lambda = \Lambda\Omega$$

$$\Rightarrow \Lambda\Omega|\omega_i\rangle = \omega_i\Lambda|\omega_i\rangle$$

$$\Omega(\Lambda|\omega_i\rangle) = \omega_i(\Lambda|\omega_i\rangle)$$

also an eigen vector with same eigen value ω_i

But we assumed Non-degeneracy.

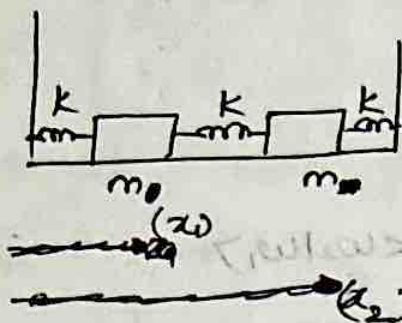
So, $\Lambda|\omega_i\rangle = \underbrace{\lambda}_{\text{constant}}|\omega_i\rangle = (\text{is only possible})$

$\Rightarrow |\omega_i\rangle$ is eigen vector, for both Λ, Ω

So, if $\Omega\Lambda = \Lambda\Omega$ and non-degenerate and Hermitian both have same set of eigen vectors.

$[\Omega, \Lambda] = \text{commutator} = \Omega\Lambda - \Lambda\Omega$

Q: Solution of a physics problem using operator and basis



$$\ddot{x}_1 = -\frac{2k}{m}x_1 + \frac{k}{m}x_2$$

$$\ddot{x}_2 = \frac{k}{m}x_1 - \frac{2k}{m}x_2$$

Solving these eqn from new techniques
finding $x_1(t)$, $x_2(t)$ given initial values.

① Writing this problem in linear vector space

$$\begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{pmatrix} = \begin{pmatrix} \Omega_{11} & \Omega_{12} \\ \Omega_{21} & \Omega_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$= \underbrace{\begin{bmatrix} -\frac{2k}{m} & \frac{k}{m} \\ \frac{k}{m} & -\frac{2k}{m} \end{bmatrix}}_{\Omega} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\Omega = \Omega^\dagger \quad (\text{Hermitian})$$

$$\ddot{\vec{x}}(t) = \Omega \vec{x}(t)$$

Assume Basis vectors in 2D as $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\text{Then, } \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

② So, need to Diagonalise the Ω

④ Change the basis vector set $\rightarrow$ another basis vector set where the matrix Ω = diagonalised

⑤ Then we have de-coupled equations

Since, Ω = Hermitian, ~~the~~

(Ω = diagonalised basis) = (basis formed by eigen vector of Ω)

⑥ step ①: $|1\rangle, |2\rangle$ are eigen vector of Ω

$$\Omega |1\rangle = \omega_1^2 \cdot |1\rangle \quad \Omega |2\rangle = \omega_2^2 \cdot |2\rangle$$

$$\Omega = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$(2 - \lambda)^2 - 1 = 0$$

$$(1 - \lambda)^2 = 0$$

$$\lambda = -1, 3$$

$$\omega_1^2 = -1, \quad \omega_2^2 = 3$$

$$\lambda = -1, -3$$

⑦

$$\begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -3 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = + \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$x_1 + x_2 = 0$$

$$x_1 = x_2$$

$$x_1 = -x_2$$

$$|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$|2\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

step ②:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = |\alpha(t)\rangle = x_1' \cdot |1\rangle + x_2' \cdot |2\rangle$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{x_1'}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \frac{x_2'}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow \begin{aligned} x_1'(0) &= \frac{x_1(0) - x_2(0)}{\sqrt{2}} \\ x_2'(0) &= \frac{x_1(0) + x_2(0)}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned} x_1' + x_2' &= \sqrt{2} x_1' \\ -x_1' + x_2' &= \sqrt{2} x_2' \end{aligned}$$

$$x_2' = \frac{x_1 + x_2}{\sqrt{2}}$$

$$x_1' = \frac{x_1 - x_2}{\sqrt{2}}$$

step 3:

$$\begin{bmatrix} \ddot{x}_1' \\ \ddot{x}_2' \end{bmatrix} = \begin{bmatrix} -\omega_1^2 & 0 \\ 0 & -\omega_2^2 \end{bmatrix} \begin{bmatrix} x_1' \\ x_2' \end{bmatrix} \quad \text{then } \omega_1^2, \omega_2^2 = 3, 1$$

$$x_1 = \frac{(x_1' + x_2')}{\sqrt{2}} \quad \frac{d^2}{dt^2} x_1 = \frac{(\ddot{x}_1' + \ddot{x}_2')}{\sqrt{2}}$$

$$x_2 = \frac{(-x_1' + x_2')}{\sqrt{2}} \quad \ddot{x}_2 = \frac{-\ddot{x}_1' + \ddot{x}_2'}{\sqrt{2}}$$

$$\ddot{x}_1' = -\omega_1^2 x_1'$$

$$x_1'(t) = x_1'(0) \cos(\omega_1 t)$$

$$\ddot{x}_2' = -\omega_2^2 x_2'$$

$$x_2'(t) = x_2'(0) \cos(\omega_2 t)$$

$$x_1 = \frac{x_1'(0) \cos \omega_1 t + x_2'(0) \cos \omega_2 t}{\sqrt{2}} \quad \begin{cases} \omega_1 = \sqrt{\frac{3k}{m}} \\ \omega_2 = \sqrt{\frac{k}{m}} \end{cases}$$

$$x_1 = \langle i, 1 | \rangle \cdot x_1'(0) \cos(\omega_1 t) + \langle i, 2 | \rangle \cdot x_2'(0) \cos(\omega_2 t)$$

$$\begin{aligned} x_1 &= \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \left(\frac{x_1(0) - x_2(0)}{\sqrt{2}} \right) \cos \omega_1 t + \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \left(\frac{x_1(0) + x_2(0)}{\sqrt{2}} \right) \cos \omega_2 t \end{aligned}$$

$$x_1 = \frac{1}{2} \left[\frac{(x_1(0) - x_2(0)) \cos \omega_1 t}{1} + \frac{(x_1(0) + x_2(0)) \cos \omega_2 t}{1} \right]$$

Step ①: Solve the eigen value problem of Ω

Step ②: Find the coeff. of $\langle 1 | x(0) \rangle = x_1(0)$

$$\langle 2 | x(0) \rangle = x_2(0)$$

$$|x(0)\rangle = x_1(0) \cdot |1\rangle + x_2(0) \cdot |2\rangle$$

Step ③: Append to each coeff $x_i(0)$ $i=1, 2$
time dependence $\cos \omega_i(t)$ to get $|x_i(t)\rangle$

$$x_2 = \langle [0, 1] | \begin{bmatrix} 1 \\ 4 \end{bmatrix} \rangle \cdot \frac{1}{\sqrt{2}} \cdot x_1(0) \cdot \cos \omega_1 t$$

$$+ \langle [0, 1] | \begin{bmatrix} 1 \\ 1 \end{bmatrix} \rangle \cdot \frac{1}{\sqrt{2}} \cdot x_2(0) \cdot \cos \omega_2 t$$

$$x_2 = \frac{1}{\sqrt{2}} \left[\frac{-(x_1(0) - x_2(0))}{\sqrt{2}} \cdot \cos \omega_1 t + \frac{(x_1(0) + x_2(0))}{\sqrt{2}} \cdot \cos \omega_2 t \right]$$

$$= \cos \omega_1 t \left(\frac{x_2(0) - x_1(0)}{2} \right) + \cos \omega_2 t \left(\frac{x_1(0) + x_2(0)}{2} \right)$$

$$\omega_1 = \sqrt{\frac{3k}{m}} \quad \omega_2 = \sqrt{\frac{k}{m}}$$

$$x_2(t) = x_2(0) \left(\frac{\cos \omega_2(t) + \cos \omega_1 t}{2} \right) + x_1(0) \left(\frac{\cos \omega_2 t - \cos \omega_1 t}{2} \right)$$

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} \frac{\cos \omega_2 t + \cos \omega_1 t}{2} & \frac{\cos \omega_2 t - \cos \omega_1 t}{2} \\ \frac{\cos \omega_2 t - \cos \omega_1 t}{2} & \frac{\cos \omega_2 t + \cos \omega_1 t}{2} \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}$$

Properties:

① Final state vector is obtained from initial state vector by multiplication with a matrix (vector)

② This matrix is independent of the initial state vector.
 → called as propagator

$$|\alpha(t)\rangle = \underbrace{U(t)}_{\text{propagator}} |\alpha(0)\rangle \quad \text{--- (1)}$$

Finding a solution $\Rightarrow$ finding the propagator

$$|\alpha(t)\rangle = |1\rangle \langle 1 | \alpha(0) \rangle \cos \omega_1 t + |1\rangle \langle 2 | \alpha(0) \rangle \cos \omega_2 t \quad \text{--- (2)}$$

From ①, ②,

components of $U(t)$ depend on eigenvector and values of operators

$$U(t) = |1\rangle \langle 1 | \cos \omega_1 t$$

$$+ |1\rangle \langle 2 | \cos \omega_2 t = \sum_{i=1}^2 |i\rangle \langle i | \cos \omega_i t$$

$$U_{11} = \langle 1 | U | 1 \rangle$$

$$= \langle 1 | \left\{ \sum_{i=1}^2 |i\rangle \langle i | \cos \omega_i t \right\} | 1 \rangle$$

$$= \sum_{i=1}^2 \langle 1 | i \rangle \langle i | 1 \rangle \cos \omega_i t$$

$$U_{11} = \frac{1}{2} (\cos \omega_1 t + \cos \omega_2 t)$$

Conclusion:

$$|\dot{x}^0\rangle = \Omega |x\rangle$$

① solve the eigen value for Ω

② construct U : $U_{11} = \langle 1|U|1\rangle$

③ $|x(t)\rangle = U|x(0)\rangle \Rightarrow$ directly write this

* Normal Mode:

ff $|x(0)\rangle = |I\rangle$

$$|x(t)\rangle = U|x(0)\rangle$$

$$= \cos \omega_1 t \cdot |I\rangle + \sin \omega_1 t \cdot |II\rangle$$

ff $|x(0)\rangle = |II\rangle$

$$|x(t)\rangle = \cos \omega_2 t \cdot |II\rangle$$

ff $t=0$ state depends only on $|I\rangle / |II\rangle$ then $|x(t)\rangle$ also depends on same.

Then

$$U_{I,II} = \begin{bmatrix} \cos \omega_1 t & 0 \\ 0 & \cos \omega_2 t \end{bmatrix}$$

* Quantum Theory

state vector : $|\psi\rangle$

$$\hbar = \frac{h}{2\pi}$$

h : Planck's const.

$$\boxed{i\hbar |\dot{\psi}\rangle = H |\psi\rangle}$$

Hamiltonian operator. ($H = H^\dagger$)

Schrodinger eqn — ①

$|\psi(t)\rangle = ?$ given $|\psi(0)\rangle$ and $|\dot{\psi}\rangle$ (as ①)

(Griffiths book) ①
(Shankar book) ②

* Problems with classical physics:

① Particles

↓
localised bundle of energy and momentum

$$q(t) \quad \dot{q}(t) \quad q_1(p)$$

these are used for eqn of motion to find trajectory of particle

✓ Many types of waves: plane, spherical.

For planar waves:

$$\psi(x,t) = A \cdot \exp \left[i \left(\frac{2\pi}{\lambda} x - \frac{2\pi}{T} t \right) \right] = A \cdot e^{i\phi}$$

↓ Amplitude
 ↓ wavelength
 ↓ time period
 ↓ phase

wave number = $k = \frac{2\pi}{\lambda}$

angular freq = $\omega = \frac{2\pi}{T}$

phase speed/velocity = $\frac{\omega}{k}$ (make $\phi = 0$)

phase $\phi = \frac{2\pi}{\lambda} x - \frac{2\pi}{T} t$

Intensity of wave $\propto |A|^2$

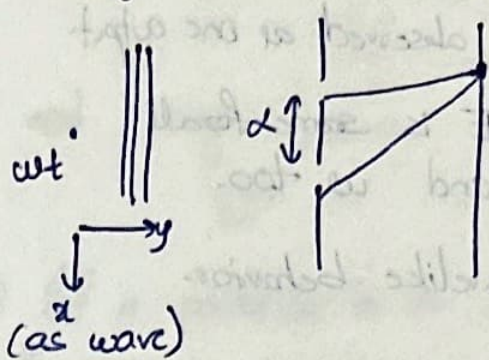
$$x = \frac{\omega}{k} t$$

const phase
move with speed $\frac{\omega}{k}$

If EM wave:

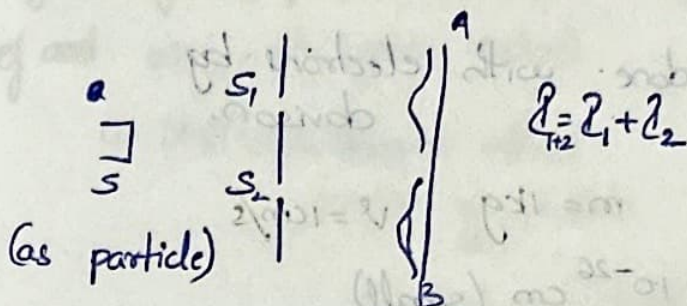
measured as momentum, Energy.

Young's Double Slit



$$\text{maxima} = (2n+1) \frac{\lambda}{2} = d_2 - d_1$$

$$\text{minima} = n\lambda = d_2 - d_1$$



⊙ The particle going through S_1 and S_2 is open, or not

⊙

When light is used as source, we observe histogram on the AB. There are detectors on AB which activate almost at same time

$$E = pc + m^2c^4 \quad \text{and} \quad m=0 \quad h = 10^{-27} \text{ erg sec} \quad (\text{Planck's const})$$

$$E = pc, \quad \hbar = \frac{h}{2\pi}$$

When both slits are open we observe interference. ie light behaves as both particle and wave.

With each photon we associate a wave fn ψ called the prob amplitude / amplitude.

$$|\psi(x)\rangle \in \text{Hilbert space}$$

$|\psi(x)|^2 \rightarrow$ prob of finding photon at x .

Each photon with Energy E and momentum p

$$\hbar \omega = E$$

$$k = \frac{p}{\hbar}$$

- ✓ This prob dist needs to be tested across various experiments and $d_{1+2} = d_1 + d_2$ also observed as one output
- ✓ If light = monochromatic, then E is same for all, and w , too.
(λ = same for all)

② De-Broglie: Each particle has wavelike behavior.

$$\lambda = \frac{h}{p}$$

Young's exp ~~was~~ was done with electrons by diffraction.

Q: Young's exp done with $m = 1 \text{ kg}$ } $v = 1 \text{ cm/s}$

$$\lambda = \frac{h}{p} = \frac{h}{mv} \approx 10^{-26} \text{ cm (small)}$$

and we will also observe $d_{1+2} = d_1 + d_2$
(for macroscopic particles)

* Postulates of QM: (non-relativistic)

Take a single particle.

Classical Mech

Quantum Mech

① State of particle at a given time defined by $x(t)$ and $p(t)$ as a point in phase space

① State of particle is represented by vector $|\psi(t)\rangle$ in Hilbert's space

② Every dynamical variable $w = w(x, p)$

② Position variables x, p are represented by Hermitian operators X, P .

x = position and p = momentum

and have the matrix element in the eigen basis of x

$$\langle x | X | x' \rangle = x \delta(x - x')$$

$$\langle x | P | x' \rangle = -i\hbar \delta'(x - x')$$

4. OM, $\delta = \text{dirac delta function}$

$$S' = \frac{\partial S}{\partial \alpha}$$

and

$$\Omega(x, p) \equiv \omega(x \rightarrow X, p \rightarrow P)$$

change variable

$\uparrow$
 $\omega(x, p)$

any dynamic variable

and any dynamic
CM

OM

③ If a particle is in state given by x, p ; measurement of ω with given $\omega(x, p)$
 $\omega(x, p) \Rightarrow$ doesn't effect position of particle.

③ If a particle is in state $|\psi\rangle$; measurement of a variable corresponding to operator Ω will yield one of the eigen values of Ω with prob.

$P(\omega) \propto |\langle \omega | \psi \rangle|^2$
④ State of the system will change from $|\psi\rangle$ to $|\omega\rangle$ as a result of measurement.
 $\omega = \text{eigen vector}$

④ State variables change with time according to Hamilton's equations.

④ The state vector $|\psi(t)\rangle$ obeys the schrodinger eqn.

$$\dot{x} = \frac{\delta H}{\delta p}$$

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$$

$$\dot{p} = -\frac{\delta H}{\delta x}$$

$H(x, p) \rightarrow H(x \rightarrow X, p \rightarrow P)$
 $\Downarrow$
Quantum Hamiltonian operator $\Downarrow$ classical Hamiltonian and change x, p to operators.

④ Hilbert space = abstract vector space and contains various state vectors like $|\psi(t)\rangle$